# QMCA Isomorphic 34 Removed

Quantum Measurement and Time Evolution

Answer the questions below. Correct answers are in RED, while incorrect answers are in BLUE.

**Q1. A particle in a one-dimensional infinite square well is prepared in the state $\sqrt{2/7}\,\psi_1(x) + \sqrt{5/7}\,\psi_3(x)$. What is the maximum value that could be measured for the energy of this system?**

**A.** $\boldsymbol{5/7}$
This is a probability, not an energy eigenvalue. It confuses the coefficient magnitude squared with a possible result of an energy measurement.

**B.** $\boldsymbol{E_1}$
$E_1$ is a possible energy outcome, but it is not the maximum possible outcome for a state containing both $\psi_1$ and $\psi_3$. This choice selects the lower eigenvalue instead of the largest allowed one.

**C.** $\boldsymbol{E_3}$
For a superposition $\sqrt{2/7}\,\psi_1 + \sqrt{5/7}\,\psi_3$, an energy measurement can return only eigenvalues whose eigenstates appear with nonzero coefficient. The possible outcomes are $E_1$ and $E_3$, and since $E_3 > E_1$ in an infinite square well, the maximum measurable energy is $E_3$.

**D.** $\boldsymbol{(2/7)E_1 + (5/7)E_3}$
This is the expectation value of energy, not a possible single-measurement result. It confuses an ensemble average with an eigenvalue outcome.

**E.** **Any $\boldsymbol{E_n}$ could be measured.**
Before any other disturbing measurement, only energy eigenstates with nonzero expansion coefficients can be measured. This choice incorrectly includes eigenvalues absent from the state.

***Answer:*** *C*

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**Q2. Suppose instead you measure the energy of the system $\sqrt{2/7}\,\psi_1(x) + \sqrt{5/7}\,\psi_3(x)$, and this time you find the most probable value for the energy. What is the normalized state of the system after this measurement?**

**A.** $\boldsymbol{\sqrt{5/7}\,\psi_3(x)}$
This is the projected component associated with the most probable outcome before renormalization. After a measurement yielding $E_3$, the state must be normalized to $\psi_3(x)$.

**B.** $\boldsymbol{\sqrt{2/7}\,\psi_1(x)}$
This is the less probable energy component and is also not normalized. It confuses the smaller-probability branch with the most probable measurement result.

**C.** $\boldsymbol{\psi_1(x)}$
$\psi_1(x)$ would be the normalized state if the measured energy were $E_1$. This choice normalizes correctly but selects the less probable branch.

**D.** $\boldsymbol{\psi_3(x)}$
The probabilities are $P(E_1) = 2/7$ and $P(E_3) = 5/7$, so the most probable energy is $E_3$. After measuring $E_3$, the projection postulate says the state collapses to the corresponding normalized eigenstate $\psi_3(x)$.

**E.** $\boldsymbol{\sqrt{2/7}\,\psi_1(x) + \sqrt{5/7}\,\psi_3(x)}$
An ideal energy measurement collapses the state to the measured energy eigenstate. This choice incorrectly assumes the measurement leaves the original superposition unchanged.

***Answer:*** *D*

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**Q3. A particle initially in the state $\sqrt{2/7}\,\psi_1(x) + \sqrt{5/7}\,\psi_3(x)$ is subjected to a high-accuracy position measurement. Immediately afterward, you measure the energy. At this point, what value(s) could you get for the energy of the particle?**

**A. Any $E_n$**
A high-accuracy position measurement prepares a strongly localized state rather than preserving the original two-term energy expansion. Such a localized state can be expanded in the complete infinite-square-well energy basis, so a later energy measurement can yield any discrete $E_n$ with nonzero expansion amplitude.

**B. $(2/7)E_1 + (5/7)E_3$**
This is the original energy expectation value, not a possible measured energy after a sharp position measurement. It ignores that the position measurement changes the state.

**C. $E_1$ only**
A localized position state is not generally the ground energy eigenstate. This choice incorrectly treats position localization as collapse into the lowest energy state.

**D. Just $E_1$ or $E_3$**
The position measurement alters the state into a localized wave packet with components in many energy eigenstates. This choice incorrectly preserves only the original two-state energy support.

**E. Any continuous energy $E$**
The infinite square well has a discrete energy spectrum. This choice confuses a localized wave packet with a free particle having continuous energies.

***Answer:*** *A*

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**Q4. Instead of measuring energy immediately after the high-accuracy position measurement, consider waiting for some time before measuring energy. Would the probability of measuring $E_4$ depend on how long you wait between the position and energy measurements?**

**A. Yes**
For a time-independent Hamiltonian, each energy-basis coefficient gains only a phase factor $e^{-iE_nt/\hbar}$. The probability $|c_4|^2$ is unchanged by that phase.

**B. No**
After the position measurement, the new state can be written as $\sum_n c_n\psi_n$. Time evolution gives $\sum_n c_n e^{-iE_nt/\hbar}\psi_n$, so the probability of measuring $E_4$ is $|c_4e^{-iE_4t/\hbar}|^2 = |c_4|^2$.

***Answer:*** *B*

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**Q5. Choose the best explanation for your answer to the previous question from the choices below.**

**A. Over time the wave function of the particle will relax back to the state it had before the position measurement.**
Unitary time evolution in a closed infinite well does not generally relax the state back to an earlier state. This introduces dissipative behavior that is not part of the model.

**B. The probability of measuring a particular energy changes because the wave function has time-dependent phases.**
The energy-basis phases change with time, but their absolute squares do not. This choice mistakes phase evolution for a change in probabilities.

**C. The probability of measuring a particular energy is time-independent.**
For a time-independent Hamiltonian, energy eigenstate coefficients evolve only by phase factors. Since probabilities are absolute squares, $|c_n e^{-iE_n t/\hbar}|^2 = |c_n|^2$, so each energy outcome probability remains constant after the new state has been prepared.

**D. Although the probability of measuring a particular energy varies in time, the expectation value of energy is time-independent.**
The energy expectation value is constant, but the statement that individual energy probabilities vary is incorrect for a time-independent Hamiltonian. This mixes a true conservation statement with a false probability statement.

**E. The position measurement would not alter the energy distribution of the system.**
A high-accuracy position measurement generally changes the energy distribution. The later constancy refers to the energy probabilities after that new post-position-measurement state is formed.

***Answer:*** *C*

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**Q6. Immediately after an energy measurement on a state $\psi(x) = C_2\psi_2(x) + C_4\psi_4(x)$, what is the state of the system?**

**A. $C_2E_2\psi_2(x) + C_4E_4\psi_4(x)$**
This is proportional to $H\psi$, not the post-measurement state. Measurement collapse selects one eigenstate rather than applying the Hamiltonian and keeping a superposition.

**B. $E_2\psi_2(x) + E_4\psi_4(x)$**
This expression applies energy factors and also drops the original coefficients. It confuses the eigenvalue equation with measurement collapse.

**C. Undetermined, but it is either $E_2\psi_2(x)$ or $E_4\psi_4(x)$**
The post-measurement state is an eigenfunction, not the eigenvalue times the eigenfunction. This choice includes incorrect energy factors in the state vector.

**D. Undetermined, but it is either $\psi_2(x)$ or $\psi_4(x)$**
An ideal energy measurement collapses the system to the normalized eigenstate associated with the measured eigenvalue. Because the initial state has components only in $\psi_2$ and $\psi_4$, the post-measurement state is either $\psi_2(x)$ or $\psi_4(x)$ depending on the random outcome.

***Answer:*** *D*

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**Q7. Immediately after an energy measurement on the state $\psi(x) = C_2\psi_2(x) + C_4\psi_4(x)$, the system is in the state $\hat{H}\psi(x)$.**

**A. True**
Applying $\hat{H}$ gives $C_2E_2\psi_2+C_4E_4\psi_4$, which is not generally the collapsed post-measurement state. This confuses operator action with the measurement projection postulate.

**B. False**
After an ideal energy measurement, the state collapses to the normalized energy eigenstate corresponding to the measured eigenvalue. The vector $\hat{H}\psi$ is generally an unnormalized superposition, so it is not the state after a single measurement.

***Answer:*** *B*

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**Q8. There is one definite energy $E$ associated with the state $\psi(x) = C_2\psi_2(x) + C_4\psi_4(x)$, where both coefficients are nonzero.**

**A. True**
A superposition of different energy eigenstates is not itself an energy eigenstate unless only one coefficient is nonzero. This choice confuses an expectation value with a definite measured value.

**B. False**
If both $C_2$ and $C_4$ are nonzero, $\psi = C_2\psi_2 + C_4\psi_4$ is not an eigenstate of $\hat{H}$. An energy measurement can yield $E_2$ or $E_4$ probabilistically, so there is no single definite energy before measurement.

***Answer:*** *B*

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**Q9. For the state $\psi(x) = C_2\psi_2(x) + C_4\psi_4(x)$, there is a certain probability that an energy measurement will yield $E_4$. If the constant $C_4$ is replaced with $C_4e^{i\alpha}$, where $\alpha$ is real and $0 < \alpha < 2\pi$, what happens to the probability that an energy measurement will result in $E_4$?**

**A. The probability will not change**
The probability of measuring $E_4$ is the squared magnitude of the coefficient of $\psi_4$, namely $|C_4|^2$. Replacing $C_4$ by $C_4e^{i\alpha}$ leaves that magnitude unchanged because $|e^{i\alpha}| = 1$.

**B. The probability will change**
A pure phase factor changes the complex phase but not the magnitude of the coefficient. This choice confuses phase changes with probability changes in the same measurement basis.

**C. The probability depends on the particular value of $\alpha$**
The angle $\alpha$ affects relative phase, but not $|C_4e^{i\alpha}|^2$. This choice incorrectly treats the phase angle as entering the energy probability directly.

***Answer:*** *A*

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**Q10. A particle is in a one-dimensional infinite square well initially in the first excited energy eigenstate. As time passes, which of the following statements describes the behavior of the wave function of the particle?**

**A. The real part will remain constant; the imaginary part is always zero.**
A stationary energy eigenstate still acquires a complex phase $e^{-iE_2t/\hbar}$. This choice ignores the time-dependent phase factor.

**B. The real part will oscillate with time; the imaginary part is always zero.**
The phase factor generally produces an imaginary part as well as a changing real part. This choice keeps only the cosine part of the phase and discards the sine part.

**C. The real and imaginary parts will both oscillate with time.**
The first excited eigenstate evolves as $\psi(x,t) = \psi_2(x)e^{-iE_2t/\hbar}$. If $\psi_2(x)$ is chosen real, then the real part is $\psi_2(x)\cos(E_2t/\hbar)$ and the imaginary part is $-\psi_2(x)\sin(E_2t/\hbar)$, so both oscillate.

**D. The real part will remain constant; the imaginary part will oscillate with time.**
The real part contains the factor $\cos(E_2t/\hbar)$, so it is not constant. This choice includes phase evolution only partly.

**E. The real and imaginary parts will both remain constant.**
The state acquires a time-dependent phase even though its probability density is stationary. This choice confuses stationary probability density with a completely time-independent complex wavefunction.

***Answer:*** *C*

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**Q11. What part(s) of the time dependent, generally complex, wave function $\psi(x,t)$ do you need in order to predict the probability of a particular outcome of a position measurement in an arbitrary interval?**

**A. Only the real part.**
For a general complex wavefunction, the probability density is $|\psi|^2 = (\mathrm{Re}\,\psi)^2 + (\mathrm{Im}\,\psi)^2$. Ignoring the imaginary part can give an incorrect probability.

**B. Only the imaginary part.**
The real part also contributes to $|\psi(x,t)|^2$. This choice is the complementary mistake of ignoring a necessary component.

**C. Either the real or imaginary part.**
Either component alone is insufficient for a general complex wavefunction. This would work only in special cases where the omitted component is known to vanish.

**D. Both the real and imaginary parts.**
Position probabilities are computed from the probability density $|\psi(x,t)|^2 = \psi^*\psi$. For a generally complex wavefunction, this equals $(\mathrm{Re}\,\psi)^2 + (\mathrm{Im}\,\psi)^2$, so both parts are needed.

**E. The wave function does not contain information about the measurement outcome.**
The wavefunction determines probabilities for position outcomes through the Born rule. This choice denies the probabilistic content of $\psi$.

***Answer:*** *D*

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**Q12. Choose the best explanation for your answer to the previous question from the choices below.**

**A. Physical observables like position are real quantities.**
Although measured positions are real, their probabilities are computed from the full complex wavefunction. This choice confuses the reality of an observable value with using only the real part of $\psi$.

**B. Only the real part carries position dependence, and the imaginary part cancels out in $\psi^*(x,t)\psi(x,t)$.**
The imaginary part can also depend on position and contributes positively to $|\psi|^2$. This choice incorrectly says the imaginary part cancels instead of contributing as $(\mathrm{Im}\,\psi)^2$.

**C. If $\psi(x,t)$ were purely real, then there would be no time dependence in $|\psi(x,t)|^2$.**
This is not the best general reason for needing both parts of a complex wavefunction. The question concerns arbitrary complex states, and position probabilities are determined by the full modulus squared.

**D. $|\psi(x,t)|^2$ can only be determined in general with both parts, because $\psi(x,t)$ might not be purely real.**
The probability density for position is $|\psi(x,t)|^2 = \psi^*(x,t)\psi(x,t)$. Since a general wavefunction may have nonzero real and imaginary parts, both are needed to evaluate the modulus squared.

**E. The wave function for a particle in a box never has an imaginary part.**
A time-dependent quantum wavefunction is generally complex, even if a spatial eigenfunction is chosen real at one instant. This choice incorrectly assumes the imaginary part is absent.

***Answer:*** *D*

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**Q13. Which of the following statements are always true for a particle in a one-dimensional infinite square well with a time-independent Hamiltonian?**

**A. The energy expectation value is time-dependent only for energy eigenstates.**
Energy eigenstates have constant energy expectation values. This choice reverses the stationary behavior of energy eigenstates.

**B. The energy expectation value is time-independent only for energy eigenstates.**
For a time-independent Hamiltonian, the energy expectation value is conserved for any state, not just eigenstates. This choice makes a true property too restrictive.

**C. The energy expectation value is always time-independent.**
In an infinite square well with a time-independent Hamiltonian, $\langle H\rangle$ is conserved. Expanding the state in energy eigenstates gives $\langle H\rangle = \sum_n |c_n|^2 E_n$, and the magnitudes $|c_n|^2$ do not change with time.

**D. The energy expectation value is always time-dependent.**
This contradicts energy conservation for a time-independent Hamiltonian. It confuses time-dependent phases in the state with a time-dependent value of $\langle H\rangle$.

**E. None of the above.**
One of the statements is true: the energy expectation value is always time-independent for this time-independent system. Therefore "none of the above" is not correct.

***Answer:*** *C*

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**Q14. Which of the following statements are always true for a particle in a one-dimensional infinite square well?**

**A. The position expectation value is time-dependent for energy eigenstates.**
Energy eigenstates in the infinite well have stationary probability densities. Their position expectation value is therefore constant in time.

**B. The position expectation value is time-independent for energy eigenstates.**
For an energy eigenstate, time evolution only adds an overall phase $e^{-iE_n t/\hbar}$. That phase cancels in $|\psi(x,t)|^2$ and in $\langle x \rangle$, so the position expectation value is time-independent.

**C. The position expectation value is always time-independent.**
Superpositions of different energy eigenstates can contain time-dependent interference terms in $|\psi(x,t)|^2$. Therefore $\langle x \rangle$ need not always be constant.

**D. The position expectation value is always time-dependent.**
Energy eigenstates provide counterexamples because their position expectation values are constant. This choice overgeneralizes from nonstationary superpositions.

**E. None of the above.**
Option B is true for energy eigenstates. This choice misses the stationary-state property.

***Answer:*** *B*

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**Q15. If you start at time $t = 0$ with wave function $|\psi_3\rangle$, does the position probability density vary with time?**

**A. Yes**
Although the state gains a time-dependent phase, the probability density is unchanged. This choice confuses phase evolution with changes in $|\psi|^2$.

**B. No**
$|\psi_3\rangle$ is an energy eigenstate, so $\psi_3(x,t) = \psi_3(x)e^{-iE_3 t/\hbar}$. The phase cancels in $|\psi_3(x,t)|^2$, leaving a time-independent position probability density.

**C. More information is needed to decide**
Knowing that the initial state is an energy eigenstate is enough. This choice overlooks the standard stationary-state result.

***Answer:*** *B*

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**Q16. If you start at time $t = 0$ with wave function $\frac{1}{\sqrt{2}}(|\psi_2\rangle + |\psi_4\rangle)$, does the position probability density vary with time?**

**A. Yes**
The two energy components acquire different phases, giving $\psi(t) = \frac{1}{\sqrt{2}}(e^{-iE_2 t/\hbar}\psi_2 + e^{-iE_4 t/\hbar}\psi_4)$. The interference term in $|\psi(x,t)|^2$ depends on $\cos((E_4 - E_2)t/\hbar)$, so the position density varies with time.

**B. No**
A superposition of different energy eigenstates is generally not stationary. This choice incorrectly applies the single-eigenstate result to a two-state superposition.

**C. More information is needed to decide**
The given superposition of two distinct energy eigenstates is enough to determine that the density generally changes in time. This choice misses the role of relative phase evolution.

***Answer:*** *A*

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**Q17. Of the choices given below, which state will have the fastest variation in time for the position probability density?**

**A.** $|\psi_7\rangle$
A single energy eigenstate has a time-independent position density. This choice has no beating frequency at all.

**B.** $\frac{1}{\sqrt{2}}(|\psi_3\rangle + |\psi_7\rangle)$
This state varies at a frequency proportional to $E_7 - E_3 \propto 49 - 9 = 40$. That is slower than the $n = 1$ to $n = 7$ superposition, whose gap is proportional to 48.

**C.** $\frac{1}{\sqrt{2}}(|\psi_2\rangle - |\psi_6\rangle)$
The minus sign changes the initial relative phase, not the beating frequency. Its energy gap is proportional to $36 - 4 = 32$, which is not the largest listed.

**D.** $\sqrt{3/10}\,|\psi_1\rangle + \sqrt{7/10}\,|\psi_7\rangle$
The time variation of a two-energy superposition is governed by the angular frequency $|E_m - E_n|/\hbar$. Since $E_n \propto n^2$ in the infinite square well, the $n = 1$ and $n = 7$ superposition has the largest listed gap, proportional to $49 - 1 = 48$, so it varies fastest.

**E. All of these states have time-independent position probability densities.**
Only single energy eigenstates have stationary position densities in this list. Superpositions of different energy eigenstates generally have time-dependent interference terms.

***Answer:*** *D*

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**Q18. Is the following statement true or false for all operators $\hat{A}$? A system which is in an eigenstate of $\hat{A}$ will stay in that state until disturbed by measurement.**

**A. True**
This is only guaranteed for special operators whose eigenstates are preserved by the Hamiltonian, such as energy eigenstates for a time-independent Hamiltonian. It is not true for an arbitrary operator $\hat{A}$.

**B. False**
Time evolution is generated by the Hamiltonian, not by an arbitrary operator $\hat{A}$. If $\hat{A}$ does not commute with the Hamiltonian or has explicit time dependence, a state that starts as an $\hat{A}$ eigenstate need not remain one.

***Answer:*** *B*

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**Q19. Choose the best explanation for your answer to the previous question from the choices below.**

**A. The system remains in an eigenstate of operator $\hat{A}$ until a measurement of a non-commuting operator is performed.**
Even without another measurement, Hamiltonian time evolution can take the state out of an $\hat{A}$ eigenstate. This choice ignores dynamical evolution.

**B. The expectation value of $\hat{A}$ is time-independent.**
The expectation value of a general operator need not be time-independent. This choice assumes conservation without requiring commutation with the Hamiltonian.

**C. The system will evolve and the state cannot be known.**
The state can be predicted by Schrödinger evolution if the Hamiltonian is known. This choice is too vague and incorrectly suggests time evolution is unknowable.

**D. The operator $\hat{A}$ may not commute with the Hamiltonian, or $\hat{A}$ may have explicit time dependence.**
A state remains an eigenstate of $\hat{A}$ under time evolution only under special compatibility conditions with the Hamiltonian. If $[\hat{A}, \hat{H}] \neq 0$ or if $\hat{A}$ explicitly depends on time, the eigenstate property need not be preserved.

**E. The operator $\hat{A}$ may not commute with the position operator.**
Commutation with position is not the relevant condition for time evolution in general. The Hamiltonian determines whether $\hat{A}$ eigenstates are preserved.

***Answer:*** *D*

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**Q20. A spin-1/2 particle is in the state $\sqrt{3/10}\,|+\rangle + \sqrt{7/10}\,|-\rangle$ in the $S_z$ basis. What is the maximum value that could be measured for $S_z$?**

**A. $7/10$**
This is a probability, not a spin angular momentum eigenvalue. It confuses the coefficient weight with a measurement outcome.

**B. $-\hbar/2$**
$-\hbar/2$ is one possible outcome but it is the smaller eigenvalue. The question asks for the maximum measurable value.

**C. $+\hbar/2$**
For a spin-1/2 measurement of $S_z$, the only possible outcomes are $+\hbar/2$ and $-\hbar/2$. The maximum of these two eigenvalues is $+\hbar/2$, independent of which outcome is more probable.

**D. $(3/10)(\hbar/2) + (7/10)(-\hbar/2)$**
This is the expectation value of $S_z$, not a single possible measurement result. It confuses an ensemble average with an eigenvalue outcome.

**E. $\sqrt{3/10}(\hbar/2) + \sqrt{7/10}(-\hbar/2)$**
Measurement outcomes are eigenvalues, and probabilities use squared magnitudes of amplitudes. This expression incorrectly weights spin values by amplitudes rather than probabilities.

***Answer:*** *C*

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**Q21. Suppose instead you measure $S_z$ of the system $\sqrt{3/10}\,|+\rangle + \sqrt{7/10}\,|-\rangle$, and this time you find the most probable value for the z-component of spin. What is then the normalized state of the system after this measurement?**

**A.** $\sqrt{7/10}\,|-\rangle$
This is the projected component associated with the most probable outcome, but it is not normalized. After measurement the state is $|-\rangle$, not the unnormalized component.

**B.** $\sqrt{3/10}\,|+\rangle$
This is the less probable branch and is also not normalized. It confuses the smaller-amplitude outcome with the most probable one.

**C.** $|-\rangle$
The probabilities are $P(+) = 3/10$ and $P(-) = 7/10$, so the most probable $S_z$ result is $-\hbar/2$. After measuring that value, the normalized post-measurement state is $|-\rangle$.

**D.** $|+\rangle$
$|+\rangle$ would be the normalized state after measuring $+\hbar/2$, the less probable outcome. This choice picks the wrong measurement branch.

**E.** $\sqrt{3/10}\,|+\rangle + \sqrt{7/10}\,|-\rangle$
An ideal measurement of $S_z$ collapses the state to an $S_z$ eigenstate. This choice incorrectly leaves the original superposition unchanged.

***Answer:*** *C*

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**Q22. Suppose you make a measurement of $S_y$ on a spin-1/2 particle. Immediately after this measurement, you then measure $S_z$. At this point, what value(s) could you get for the z-component of spin of the particle?**

**A.** $(7/10)(-\hbar/2) + (3/10)(+\hbar/2)$
This is an expectation-value-style expression, not a list of possible single outcomes of an $S_z$ measurement. It confuses averages with eigenvalue outcomes.

**B.** $-\hbar/2$ **only**
A prior $S_y$ measurement does not make $S_z$ have only one possible outcome. This choice incorrectly assumes certainty for one $S_z$ value after measuring an incompatible component.

**C.** **Just** $-\hbar/2$ **or** $+\hbar/2$
A measurement of $S_z$ can yield only the allowed discrete $S_z$ eigenvalues. For a spin-1/2 system these are $+\hbar/2$ and $-\hbar/2$, even after a prior measurement of the incompatible component $S_y$.

**D.** $+\hbar/2$**, 0, or** $-\hbar/2$
A spin-1/2 $S_z$ measurement has two outcomes, not three. This choice confuses the system with a spin-1 spectrum.

**E.** **Any continuous angular momentum value between** $-\hbar/2$ **and** $+\hbar/2$**.**
Spin component measurements have discrete eigenvalues, not a continuous range. This choice treats spin angular momentum like a classical continuous variable.

***Answer:*** *C*

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**Q23. Instead of an immediate re-measurement of $S_z$, consider waiting between the $S_y$ and $S_z$ measurements. If the Hamiltonian commutes with $S_z$, would the probability of measuring $S_z$ to be $+\hbar/2$ depend on how long you wait between the measurements?**

**A. Yes**
If $S_z$ commutes with the Hamiltonian, the magnitudes of the $S_z$-basis coefficients are time-independent. Waiting changes phases, not the probability of an $S_z$ outcome.

**B. No**
After an $S_y$ measurement, the state can be expanded in the $S_z$ basis. Under a Hamiltonian that commutes with $S_z$, the $S_z$ coefficients acquire only phase factors, so their absolute squares and hence the $S_z$ probabilities are time-independent.

***Answer:*** *B*

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**Q24. Choose the best explanation for your answer to the previous question from the choices below.**

**A. Over time the particle will relax to the spin state it had before the $S_y$ measurement.**
Closed-system spin evolution is unitary and does not generally relax to the initial state. This choice introduces dissipation that is not part of the setup.

**B. Because it is precessing, the probabilities of $S_z$ must change with time.**
Precession can change probabilities for components transverse to the conserved axis, but if the Hamiltonian commutes with $S_z$, the $S_z$ probabilities are constant. This choice applies the wrong intuition to the stated component.

**C. $S_z$ commutes with the Hamiltonian, so its probabilities are time-independent.**
When $[S_z, H] = 0$, the Hamiltonian does not mix the $S_z$ eigenspaces. Thus the probabilities for $S_z$ outcomes remain fixed in time, even though relative phases may evolve.

**D. All possible measurement probabilities in quantum mechanics are time independent because you take the absolute square of the coefficient.**
Only probabilities in a basis whose populations are not mixed by the Hamiltonian are guaranteed constant. This choice overgeneralizes the phase-cancellation argument.

**E. Once you measure $S_y$ the state is no longer in an eigenstate of the Hamiltonian.**
Being outside a Hamiltonian eigenstate can lead to phase evolution, but it does not by itself imply changing $S_z$ probabilities. The key issue is whether $S_z$ commutes with the Hamiltonian.

***Answer:*** *C*

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**Q25. Immediately after a measurement of $S_z$ on a state $|\psi\rangle = C_+|+\rangle + C_-|-\rangle$, what is the state of the system?**

**A. $C_+(\hbar/2)|+\rangle - C_-(\hbar/2)|-\rangle$**
This is proportional to applying $\hat{S}_z$ to the state, not the normalized post-measurement state. Measurement collapses to one eigenstate rather than producing this superposition.

**B. $(\hbar/2)|+\rangle - (\hbar/2)|-\rangle$**
This expression multiplies basis kets by eigenvalues and omits the original amplitudes. It is not the result of a projective measurement.

**C. Undetermined, but it is either $(\hbar/2)|+\rangle$ or $-(\hbar/2)|-\rangle$**
The state after measurement is a normalized eigenket, not the eigenvalue times the eigenket. This choice includes inappropriate eigenvalue factors in the state.

**D. Undetermined, but it is either $|+\rangle$ or $|-\rangle$**
An ideal $S_z$ measurement collapses the system into the normalized eigenstate corresponding to the measured result. Since the possible $S_z$ eigenstates are $|+\rangle$ and $|-\rangle$, the post-measurement state is one of those two.

***Answer:*** *D*

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**Q26. Immediately after a measurement of $S_z$ on the state $|\psi\rangle = C_+|+\rangle + C_-|-\rangle$, the system is in the state $\hat{S}_z|\psi\rangle$.**

**A. True**
Applying $\hat{S}_z$ gives a generally unnormalized superposition with eigenvalue factors. That is not the same as collapse to one measured eigenstate.

**B. False**
A projective measurement of $S_z$ leaves the system in the normalized eigenstate associated with the measured eigenvalue. The vector $\hat{S}_z|\psi\rangle$ is not generally normalized and does not represent a single measurement outcome.

***Answer:*** *B*

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**Q27. There exists a single definite value for the z-component of spin associated with the state $|\psi\rangle = C_+|+\rangle + C_-|-\rangle$, where both coefficients are nonzero.**

**A. True**
A superposition of $S_z$ eigenstates is not generally itself an $S_z$ eigenstate. This choice confuses a probability distribution or expectation value with a definite pre-existing outcome.

**B. False**
If both $C_+$ and $C_-$ are nonzero, the state is not an eigenstate of $S_z$. A measurement can yield $+\hbar/2$ or $-\hbar/2$ probabilistically, so no single definite $S_z$ value is associated with the state.

***Answer:*** *B*

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**Q28. For the state $|\psi\rangle = C_+|+\rangle + C_-|-\rangle$, there is a certain probability that an $S_z$ measurement will yield $-\hbar/2$. If the constant $C_-$ is replaced with $C_-e^{i\beta}$, with $\beta$ real and $0 < \beta < 2\pi$, what happens to the probability that a measurement of $S_z$ will result in $-\hbar/2$?**

**A. The probability will not change**
The probability of measuring $-\hbar/2$ is $|C_-|^2$. Multiplying $C_-$ by $e^{i\beta}$ does not change its magnitude, since $|e^{i\beta}| = 1$.

**B. The probability will change**
A phase factor changes the complex phase, not the magnitude of the coefficient. This choice confuses phase with probability in the $S_z$ basis.

**C. The probability depends on the particular value of $\beta$**
The probability in the $S_z$ basis contains $|C_-e^{i\beta}|^2$, which is independent of $\beta$. This choice incorrectly makes the phase angle directly observable in this measurement probability.

***Answer:*** *A*

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**Q29. In a three-outcome quantum toy model, an experimenter measures the flavor of particles and then immediately sends all particles that measured vanilla through the same FlavorBox again. What are the possible results of this second measurement?**

**A. The only possible measurement is vanilla with a 100% probability of measurement.**
Immediately after a flavor measurement yielding vanilla, the state is the vanilla eigenstate $|v\rangle$. Repeating the same ideal measurement immediately gives the same eigenvalue with probability 1.

**B. Vanilla, chocolate, and strawberry can be measured with equal probabilities.**
Equal probabilities would require a special superposition in the flavor basis, not the post-measurement vanilla eigenstate. This choice ignores collapse from the first flavor measurement.

**C. Vanilla, chocolate, and strawberry can be measured, but their associated probabilities cannot be determined from the information given.**
The first measurement result provides enough information: the selected particles are in the vanilla eigenstate. This choice overlooks the determinism of an immediate repeated measurement of the same observable.

**D. There is not enough information to answer this question.**
For an ideal measurement, the post-measurement state after obtaining vanilla is known. Therefore the immediate repeated FlavorBox result is determined.

***Answer:*** *A*

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**Q30. 1200 particles prepared in the flavor state $|v\rangle$ are immediately run through a TextureMeter, which measures the smooth eigenvalue 300 times, the rough eigenvalue 600 times, and the grainy eigenvalue 300 times. Which one of the following could be a valid representation of the $|v\rangle$ state in the "texture" basis?**

**A. $300|s\rangle + 600|r\rangle + 300|g\rangle$**
Measurement counts are not probability amplitudes. This state is not normalized and would give probabilities proportional to the squares of 300, 600, and 300.

**B. $\frac{1}{4}|s\rangle + \frac{1}{2}|r\rangle + \frac{1}{4}|g\rangle$**
This uses probabilities as amplitudes rather than square roots of probabilities. Its squared amplitudes would be 1/16, 1/4, and 1/16, not 1/4, 1/2, and 1/4.

**C. $\frac{1}{2}e^{i\phi_s}|s\rangle + \frac{1}{\sqrt{2}}e^{i\phi_r}|r\rangle + \frac{1}{2}e^{i\phi_g}|g\rangle$**
The observed probabilities are $300/1200 = 1/4$, $600/1200 = 1/2$, and $300/1200 = 1/4$. Therefore the amplitude magnitudes must be $1/2$, $1/\sqrt{2}$, and $1/2$, with arbitrary phases that do not affect the texture measurement probabilities.

**D. $\frac{1}{4}e^{i\phi}(|s\rangle + \sqrt{2}|r\rangle + |g\rangle)$**
The square-root ratios are present, but the normalization factor is wrong. The normalized version would require amplitudes $1/2$, $1/\sqrt{2}$, and $1/2$, not an overall factor of $1/4$.

**E. None of the above.**
Option C gives a valid normalized state with squared amplitudes matching the observed frequencies. Therefore "none of the above" is not correct.

***Answer:*** *C*

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**Q31. 800 rough particles are immediately run through the FlavorBox and then immediately run through the TextureMeter without looking at which flavors were produced. If "flavor" and "texture" commute, what can you say about the results from the TextureMeter measurements?**

**A. All three "texture" eigenvalues could be measured.**
If flavor and texture commute, the flavor measurement does not disturb the definite texture value. This choice treats the intermediate measurement as if it randomized texture.

**B. Most but not all of the particles will be measured as rough.**
Commuting observables can be measured without disturbing the already definite texture eigenvalue. The result should not merely be likely rough; it should be certain rough.

**C. All particles will be measured as rough.**
If flavor and texture commute, they have compatible eigenstates and measuring flavor does not disturb a texture eigenstate. Therefore particles prepared as rough will still be measured as rough with probability 1 in the subsequent TextureMeter measurement.

**D. The results are probabilistic and cannot be predicted.**
The commutation condition provides enough information to predict the texture result. This choice ignores the compatibility of the two observables.

**E. The "flavor" measurement will affect the "texture" measurement.**
A measurement of a commuting observable need not disturb the eigenvalue of the original observable. This choice describes noncommuting observables, not the stated case.

***Answer:*** *C*

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**Q32. True or False: Applying the Hamiltonian to a quantum state gives information about how that state changes with time.**

**A. Always true.**
The Hamiltonian is the generator of time evolution through the time-dependent Schrödinger equation $i\hbar\partial_t|\psi\rangle = \hat{H}|\psi\rangle$. Therefore applying $\hat{H}$ gives information about the time derivative of the state and hence about its evolution.

**B. True only for energy eigenstates.**
Special simplifications occur for energy eigenstates, but the time-dependent Schrödinger equation applies to arbitrary states. This choice confuses easy-to-solve cases with the general role of the Hamiltonian.

**C. Always false.**
This contradicts the central dynamical equation of quantum mechanics. The Hamiltonian directly determines time evolution through the Schrödinger equation.

***Answer:*** *A*

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**Q33. Choose the best explanation for your answer to the previous question from the choices below.**

**A. The Hamiltonian describes the time evolution of the system through the time-dependent Schrödinger equation.**
The time-dependent Schrödinger equation is $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$. Thus the Hamiltonian determines how the quantum state changes in time, not merely what energy values can be measured.

**B. The Hamiltonian always makes $\psi(x)$ oscillate with one single frequency.**
Energy eigenstates acquire single oscillatory phase factors, but a general state can involve many components and phases. The general reason is the time-dependent Schrödinger equation, not a single-frequency rule.

**C. The Hamiltonian only measures the energy at the initial time.**
The Hamiltonian is associated with energy, but applying it is not the same as making a measurement. This choice also ignores its role as the generator of time evolution.

**D. The Hamiltonian has no time dependence and therefore only lists the possible energy eigenvalues.**
Even a time-independent Hamiltonian determines the full time evolution of states. This choice reduces the Hamiltonian to a spectrum provider and misses the dynamical equation.

**E. The Hamiltonian collapses the system, making later energy measurements impossible to predict.**
Applying the Hamiltonian in the Schrödinger equation is not a measurement and does not collapse the state. This choice confuses unitary evolution with projective measurement collapse.

***Answer:*** *A*